

SRIJEET GUHA

Pune, Maharashtra, India

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Education

Birla Institute of Technology and Science - Pilani Campus

Aug. 2019 – May 2023

B.E. (Hons.) in Electrical and Electronics Engineering : CGPA - 8.18 / 10

Pilani, India

Thesis: *DynaRapid: A Novel Approach to Rapid FPGA Implementation Design Flow*

Advisors: Paolo Ienne, Andrea Guerrieri, Pankaj Arora

Technical Skills & Courses

EE Courses: Embedded Systems, Computer Architecture, Microprocessor Interfacing, Digital Design, Internet of Things

CS Courses: Data Structures and Algorithms, Object Oriented Programming, Operating Systems, Discrete Mathematics

Programming Languages: C/C++, JAVA, Python, CUDA, Assembly, Bash, Verilog

Developer's Tools: Vivado, Quartus Prime, Vitis HLS, Intel HLS, Icarus Verilog, RapidWright, Nsight Compute, Gem5

Publications

(C: Conferences & Journals, *: Equal Contribution)

[C4] **Srijeet Guha**, Andrea Guerrieri. [Precision Unwound: Fine-Tuning Loop Unrolling for Energy-efficient FPGA-based PQC using HLS](#). *In proceedings of ISQED 2025, San Francisco, USA*

[C3] **Srijeet Guha**, Andrea Guerrieri. [Iterative Frequency Tuning Targeting Energy Efficiency Ratio for FPGA-based Post-Quantum Cryptographic Cores](#). *In proceedings of ICECS 2024, Nancy, France*

[C2] Andrea Guerrieri*, **Srijeet Guha***, Chris Lavin, Eddie Hung, Lana Josipović, and Paolo Ienne. [DynaRapid: Fast-tracking from C to routed circuits](#). *In proceedings of FPL 2024, Torino, Italy*. **Best Paper Award**

[C1] Andrea Guerrieri*, **Srijeet Guha***, Lana Josipović, and Paolo Ienne. [DynaRapid: From C to FPGA in a few seconds](#). *In proceedings of FPGA 2024, Monterey, California*

Research Experiences

Adaptive Heterogeneous Systems Laboratory, HES-SO Valais-Wallis

Jan. 2024 – Dec. 2024

Research Assistant | Project Advisor: Prof. Andrea Guerrieri

Sion, Switzerland

- Developed an **iterative frequency tuning framework** to extract the best quality energy-efficient design for FPGA-based post-quantum cryptographic cores, converging to the best design **2.89×** faster than the classical approach
- Developed **precision unwound** where we explore and fine-tune high-level design parameters like loop unrolling, pipelining and dataflow optimization to reduce latency, area and energy consumption of FPGA-based PQC accelerators

Processor Architecture Laboratory, École polytechnique fédérale de Lausanne **Feb. 2023 – Dec. 2023**

Research Assistant | Project Advisor: Prof. Paolo Ienne and Prof. Andrea Guerrieri

Lausanne, Switzerland

- Developed **DynaRapid** to decrease compilation time of fully-legal placed and routed kernel designs from high-level C/C++ codes for commercial FPGAs with minimal degradation in operating frequency and increase in FPGA resources
- Reduced the C to FPGA implementation time by **33×** with a minimal degradation of **20%** in the operating frequency
- Generated a library of pre-synthesized, placed and routed building blocks to be stitched at runtime using **RapidWright**

Edinburgh Architecture and Systems Lab, The University of Edinburgh

Sept. 2022 – Dec. 2022

Research Intern | Project Advisor: Prof. Boris Grot and Dr. David Schall

Edinburgh, United Kingdom

- Improved **T2** by expanding its stride-prefetcher table to incorporate at most 3 stable strides for all loop instructions, improving prefetch coverage by **56%** while reducing accuracy by only **7%**, when tested for ARM processors on **Gem5**
- Developed **benchmarks** to measure accuracy, performance and power consumed by prefetchers implemented on **Gem5**
- Evaluated appropriate **warm-up interval** of **idle instances** for large caches with high accuracy in serverless systems

On-Board Computing Subsystem, Team Anant

Feb. 2020 – Nov. 2022

Research Engineer | Team Advisor: Prof. Meetha V. Shenoy

Pilani, India

- Designed architectures and protocols to facilitate **inter-processor** communication and data transfer on the **satellite**
- Implemented **CCSDS122 2-D** compression algorithm on the Zynq-7000 FPGA to achieve a compression ratio of 67%
- Developed **FSM-based** mode-switching algorithms with advanced graph-traversal methodologies to reduce latency and energy consumption by **2.7%** and **11%** respectively, while switching between different modes of operation of the satellite

Industrial Experiences

DevTools PerfWorks Team, [NVIDIA Corporation](#)

Jan. 2024 – Present

System Software Engineer | [Employment Letter](#)

Pune, India

- Improved data eviction policies of **L1 caches** for **global memory loads** to reduce cache miss rates by **3-4%**
- Implemented data compaction-decompaction units between the L2 cache and L1 caches of each SM to reduce XBAR congestion and data transfer latencies on Blackwell GPUs
- Introduced new software counter libraries and metrics to monitor **data bank conflicts** in L1 caches and data transfers between SM-SM **distributed shared memories** during data loads and stores
- Improved basic block optimization techniques in SASS Patching library of Blackwell GPUs to reduce profiling latency of software metrics by **37%** and GPU memory footprint by **12%**

IOSXE Polaris Team, [Cisco Systems Private Ltd.](#)

Aug. 2023 – Dec. 2023

Software Engineer | [Employment Letter](#)

Bangalore, India

- Improved the **IOSd build system** to reduce build time of bootable images by **14%**, thus improving turnaround time
- Implemented novel message-passing protocols on stack interconnect to reduce congestion in multi-processor router stacks
- Developed low-level driver software to improve communication between **IOSd SHIM** layers and the router hardware
- Introduced tests to verify the reason and timing of previous unexpected reloads in **multi-processor router stacks**

SWIFT Automatic Unit Testing Team, [Cisco Systems Private Ltd.](#)

May 2022 – Aug. 2022

Technical Undergraduate Intern | [Project Documentation](#)

Bangalore, India

- Used Software Integration and Functional Test Framework and Orchestrator Infrastructure to execute Automatic Unit Test cases and remove memory leaks from the CISCO Inter-networking Operating System (IOS)
- Simplified IOS debugging by integrating **Undo** live-recorder in **host** unit-test **pre-commit** build system of the codebase
- Owing to my performance during the internship, I was asked to join Cisco as a full-time engineer after my graduation

On-board Computing Team, [Team Robocon](#)

Nov. 2019 – March 2020

Computer Engineer

Pilani, India

- Built a robot mouse to find the shortest path using **Dijkstra's** algorithm and traverse through it using **PID** Algorithms
- Developed a **Bluetooth-based communication system** to transfer real-time position, housekeeping data, attitude, and images from drones to the user or other drones of the drone - cluster for coordination

Presentations

- [2] [Precision Unwound: Fine-Tuning Loop Unrolling for Energy-efficient FPGA-based PQC using HLS](#) at the 26th IEEE International Symposium on Quality Electronic Design, USA, March '25
- [1] [Iterative Frequency Tuning Targeting Energy Efficiency Ratio for FPGA-based Post-Quantum Cryptographic Cores](#) at the 31st IEEE International Conference on Electronics, Circuits and Systems, France, November '24

Awards & Honors

[Michal Servit Best Paper Award](#)

Sept. 2024

The 34th IEEE International Conference on Field-Programmable Logic and Applications

Torino, Italy

- Awarded for our publication on [DynaRapid](#) [C2] at the FPL'24 conference

Projects

[Cold Storage Management System](#) | Project Advisor: Prof. Meetha Shenoy

May 2022

- Interfaced **DHT11** (humidity sensor), infrared flame sensors, **MQ-3** (gas sensor) and **L293D** (motor driver), with a **STM32F407IG** processor to measure, maintain, and relay vital health parameters of the cold-storage
- Used 2 STM32F407IG processors connected using **Zigbee modules** to relay real-time information amongst each other

[16-bit RISC Processor with Floating Point ALU](#) | Project Advisor: Prof. Karri Babu Ravi Teja

May 2022

- Implemented a **5-stage** multi-cycle data path and a **21-stage** FSM based control path with multiple addressing modes
- ALU included **ripple carry** adders, **barrel shifters**, **radix 4 booth's** multiplier and logic for under/overflow detection

[Using Memory Traces to draw Insights](#) | Project Advisor: Prof. Vinay Chamola

Dec. 2021

- Developed applications to dump the **SRAM traces** on the serial monitor without altering the SRAM, thus remaining **undetected** by other applications running on the processor

[MOS Circuits using LTSpice and Microwind](#) | Project Advisor: Prof. Anu Gupta

Dec. 2021

- Designed a two-stage single-ended OPamp (folded cascode [folded amplifier + common gate stage] + gain stage) with a gain > **90 dB**, Unity Gain Bandwidth > **1 MHz**, and phase margin $\approx$ **60 degrees**

Leadership

System Engineer | Team Anant

Aug. 2021 – Aug. 2022

- Coordinated between subsystems and the Indian Space Research Org to resolve inter-subsystem dependencies and issues
- Collaborated with the subsystems, stakeholders, and the institute to finalize the annual team budget for FY' 22-23

Test Scores

GRE General Test: 322 / 340 [Verbal Reasoning: 152/170, Quantitative Reasoning: 170/170, Analytical Writing: 4.0/6.0]
TOEFL iBT: 105 /120 [Reading: 26/30, Listening: 23/30, Speaking: 28/30, Writing: 28/30]

References

Tushar Kashalikar | Engineering Manager, DevTools PerfWorks Team, NVIDIA Corporation

Email: tkashalikar@nvidia.com

Linkedin: <https://www.linkedin.com/in/tushar-kashalikar-3a958110/>

Location: Pune, Maharashtra, India

Dr. Paolo Ienne | Professor, École polytechnique fédérale de Lausanne

Email: paolo.ienne@epfl.ch

Administrative Assistant: Chantal Schneeberger (chantal.schneeberger@epfl.ch)

Website: <https://people.epfl.ch/paolo.ienne>

Location: Lausanne, Vaud, Switzerland

Dr. Andrea Guerrieri | Associate Professor, Haute Ecole Spécialisée de Suisse Occidentale

Email: andrea.guerrieri@hevs.ch

Website: <https://www.hevs.ch/en/collaborateurs/guerrieri-207310>

Location: Valais-Wallis, Sion, Switzerland

Dr. Meetha V. Shenoy | Associate Professor, Birla Institute of Technology and Science, Pilani

Email: meetha.shenoy@pilani.bits-pilani.ac.in

Website: <https://www.bits-pilani.ac.in/pilani/meetha-v-shenoy/>

Location: Pilani, Rajasthan, India